

Wireless Transmission:

Physical Layer Aspects and Radio Channel Characteristics

Radio Frequencies and Signals:

- 1) What are the wavelengths of the different band used by GSM and WLAN?

GSM:

For 900MHz Band: Wavelength Range Approximately 33.7 cm to 31.3 cm

For 1800 MHz Band Wavelength Range Approximately 17.5 cm to 16.0 cm

WLAN:

For 2.4 GHz band 12.5 cm to 12.0 cm

For 5 GHz band 6.0 cm to 5.7 cm

- 2) Which bands typically provide better penetration of walls?

(VHF) very high frequency /(UHF) ultra high frequency provides better penetration walls because of their small pathloss.

- 3) What's the disadvantage of using higher frequencies for communications? What's the advantage?

Disadvantages:

Propagation loss, Equipment cost, Interference, power consumption, less penetration, less distance covered, repeated infrastructure.

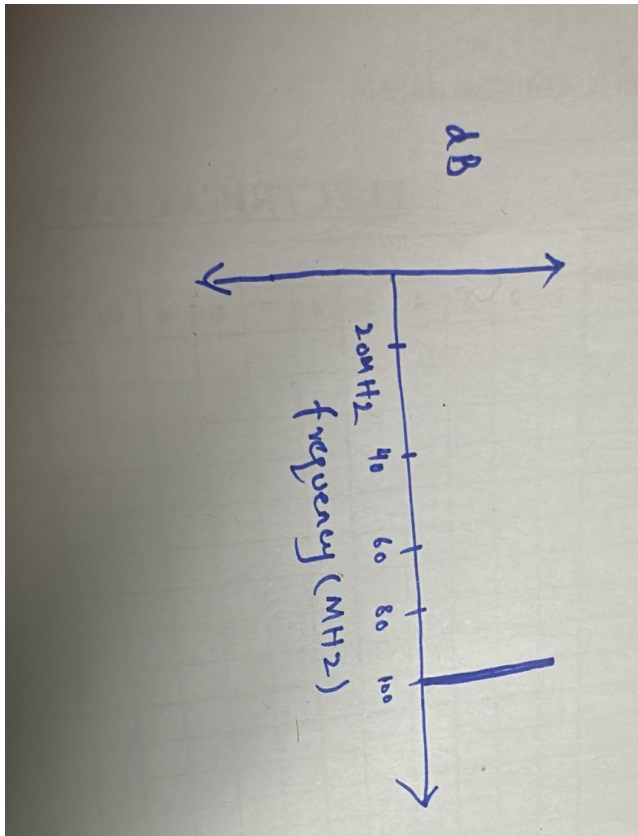
Advantages:

Greater Bandwidth, less congestion, smaller Antennas, High data rate.

- 4) Sketch the frequency spectrum of a 100 MHz sinus signal and of a symmetric rectangular signal with the same period.

Frequency spectrum of a 100 MHz sinus signal:

A sinusoidal signal with a frequency of 100 MHz has a single peak at 100 MHz in the frequency domain.

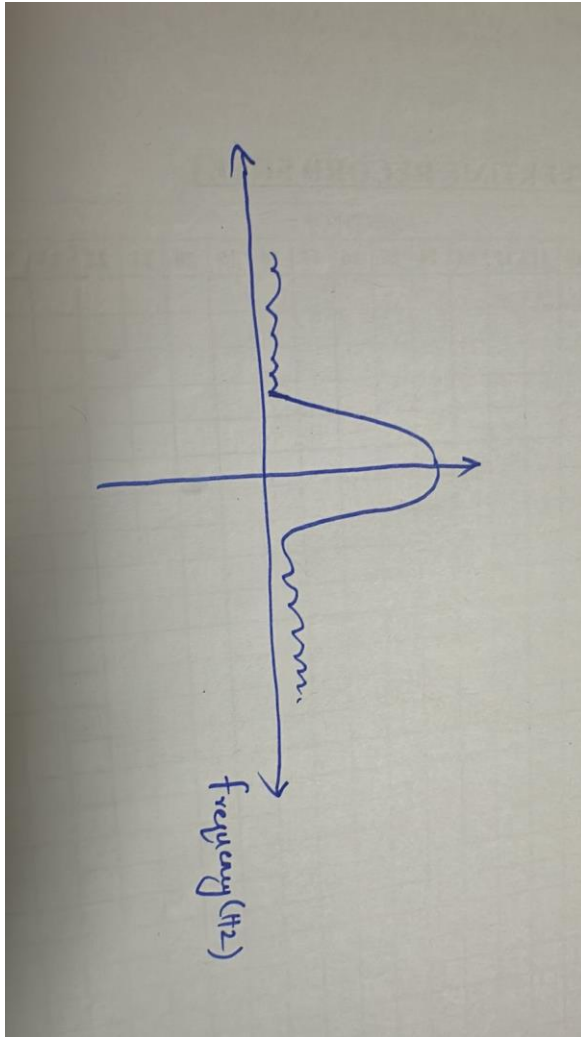


Frequency Spectrum of a 100 MHz symmetric rectangular signal:

A symmetric rectangular signal with the same period as the 100 MHz sinusoid will have harmonics at odd multiples of the fundamental frequency.

The fundamental frequency of the rectangular signal will also be 100 MHz.

The spectrum will consist of peaks at 100 MHz, 300 MHz, 500 MHz, and so on, with decreasing amplitude as the harmonic number increases.



Signal Propagation:

- 1) Why is a pathloss proportional to d^2 is assumed for line-of-sight connections? Why is d^2 considered as the best case?

In wireless communication, path loss refers to the attenuation of signal strength as it propagates through the medium, typically air, between the transmitter and receiver. The path loss is often characterized by the free-space path loss model, which assumes that the signal spreads out uniformly in three-dimensional space, leading to a decrease in signal strength with distance.

The path loss model assumes that the power density of the signal decreases as the inverse square of the distance (d^2) from the transmitter. This means that if you double the distance from the transmitter, the power density of the signal at the receiver decreases by a factor of

four. This relationship arises due to the spreading of energy over a larger area as the signal propagates outward from the source.

When line-of-sight (LOS) connections are assumed:

- **Direct Path:** In LOS scenarios, the signal travels directly from the transmitter to the receiver without significant obstructions. This direct path is the most efficient propagation scenario, as it minimizes the amount of energy lost due to absorption, reflection, and diffraction by obstacles.
- In LOS conditions, the signal spreads out uniformly in all directions from the transmitter. Since there are no obstacles to block or scatter the signal, it maintains its spherical propagation pattern. As a result, the power density of the signal decreases with the square of the distance (d^2) from the transmitter.

Consideration of d^2 as the best case in LOS scenarios is based on several factors:

- **Minimal Obstructions:** LOS connections imply that there are minimal obstacles between the transmitter and receiver, resulting in less attenuation of the signal.
- **Predictable Attenuation:** In free-space conditions, the path loss is primarily determined by the distance between the transmitter and receiver, making it relatively predictable and consistent.
- **Simplicity:** The d^2 path loss model provides a simple and effective way to estimate signal strength in LOS scenarios, facilitating system design and optimization.

However, it's essential to note that while d^2 path loss is a useful approximation in many situations, real-world propagation conditions may introduce additional factors such as multipath fading, terrain effects, and environmental obstacles, which can affect signal propagation and require more complex models for accurate prediction.

2) What are the reasons for fast fading? What is exactly happening?

Fast fading refers to rapid changes in the strength or quality of a wireless signal over short periods of time. It happens because the signal travels through different paths and gets mixed up as it reaches the receiver. This mixing causes the signal's strength to fluctuate quickly, which can affect the reliability of wireless communication.

Fast fading occurs due to rapid fluctuations in the received signal strength over short periods of time. It is primarily caused by multipath propagation, where multiple copies of the transmitted signal reach the receiver via different paths due to reflections, diffraction, and scattering in the propagation environment.

Here is what is exactly happening:

Multipath propagation, Phase differences, doppler effect, signal combining and time variations.

Quick changes in received power , Mobility + Variable channel Characteristics.

3) Why does fast fading depend on mobility? Why is fast fading frequency dependent?

Fast fading depends on mobility primarily because mobility introduces changes in the relative positions of the transmitter, receiver, and reflecting objects in the environment.

Fast fading happens because signals bounce around and mix up as they travel from the transmitter to the receiver. This mixing causes the signal's strength to quickly go up and down, making it harder for devices to communicate reliably.

Fast fading can be different depending on the frequency of the signal. Higher-frequency signals change more quickly than lower frequency ones, so they experience faster fading.

Mobility gives rise to change in paths , phases and delays the variation. Fast fading gives rise to doppler shift and is frequency dependent.

4) What is the difference between fading due to simple phase shifts between arriving signals and inter symbol interference?

Fading due to Simple Phase Shifts:

Cause:

This type of fading occurs when multiple copies of a transmitted signal arrive at the receiver via different paths. Each of these paths may experience different phase shifts due to differences in propagation distance and reflection angles.

Effect:

The phase shifts cause variations in the combined signal's amplitude at the receiver. When the phases align constructively, the received signal strength increases, but when they align destructively, the received signal strength decreases. These fluctuations in signal strength, known as fading, can lead to fluctuations in the received signal quality and may cause communication errors.

Intersymbol Interference:

Cause:

ISI occurs when symbols transmitted in a digital communication system overlap in time at the receiver due to channel dispersion or insufficient symbol spacing. In other words, the effects of previous symbols overlap with the current symbol being detected.

Effect:

ISI distorts the received signal, making it more challenging for the receiver to correctly interpret the symbols. This distortion can lead to errors in symbol detection, especially in high-speed communication systems where symbols are transmitted rapidly. ISI can be particularly problematic in systems with frequency-selective fading channels, where different frequency components of the signal experience different delays and attenuations.

Fading due to simple phase shifts results from variations in the amplitude of the received signal caused by differences in propagation paths and phase shifts between the arriving signals.

Intersymbol interference results from symbols overlapping in time at the receiver, leading to distortion and difficulties in symbol detection. It is caused by channel dispersion or insufficient symbol spacing.

While fading affects the amplitude of the received signal, ISI affects the temporal structure of the received signal by causing symbol overlap. Both phenomena can degrade the performance of communication systems, especially in wireless or high-speed digital communication environments.

5) What are the factors of the received signal which are important for error-free reception?

Signal Strength, Signal to Noise Ratio(SNR) , Channel conditions, Modulation Schemes, Error Connection Coding, Equalization, Receiver sensitivity and selectivity, Interference rejection, antenna characteristics.

Antennas:

1) What are the characteristics of an isotropic radiator?

An isotropic radiator radiates electromagnetic energy uniformly in all directions, forming a spherical radiation pattern. In other words, the radiated power density is the same in all directions around the antenna.

2) What directional effects show simple practical antennas?

Simple practical antennas exhibit directional effects such as beamwidth, directivity, gain, side lobes, back lobes, and polarization. These characteristics determine the antenna's radiation pattern and its ability to transmit or receive electromagnetic signals effectively in specific directions.

They show directive effects vertical or horizontal.

3) How can you build a directional antenna from 2 simple $\lambda/2$ antennas? What are the prerequisites?

To build a directional antenna from two simple half-wavelength ($\lambda/2$) antennas, you can use a technique called antenna arraying or antenna phasing.

Pre Requisites:

Identical Antennas, Proper Spacing, Phasing Network, feed line.

Steps to Build:

Mounting, Phasing Adjustment, Testing Optimization.

Channel Capacity – Signal vs. Interference:

1) How do interference and the received signal strength determine the capacity of a channel?

Channel capacity refers to the maximum rate at which information can be reliably transmitted over the channel.

As interference increases, the channel capacity decreases because it becomes more challenging to distinguish the desired signal from the interference.

When the received signal strength is low, the likelihood of errors increases, limiting the achievable data rate and reducing the channel capacity.

Interference reduces channel capacity by introducing errors, while higher received signal strength improves capacity by enabling more reliable communication at higher data rates.

2) Why is interference – or more accurately its absence – as important as the signal itself?

Interference, or the absence of interference, is as important as the signal itself because it directly impacts the quality and reliability of communication.

Interference can degrade the quality of the received signal, leading to errors in data transmission.

Interference limits the spectral efficiency of communication channels by reducing the achievable data rates.

Capacity Limitations, Resource Allocation, Quality of Experience.

3) What defines an interference-limited system? What happens there? How can we increase the system capacity in these systems? And how not?

An interference-limited system is characterized by the presence of significant interference that imposes constraints on the system's capacity and performance.

In interference-limited systems, the interference reduces the signal-to-noise ratio (SNR), leading to errors in data transmission. This limits the achievable data rates and spectral efficiency of the system, resulting in suboptimal performance and reduced capacity

We can increase the system capacity by Interference mitigation, power control, Adaptive modulation and coding, and spatial diversity.

To limit the increase in system capacity we need interference floor and spectrum scarcity.

Multiplexing – Splitting up the Radio Resource

1) What is the purpose of multiplexing?

Organizations implement multiplexing on their networks for two reasons:

- To enable network devices to communicate with each other without needing a dedicated connection between each device pair, although multiplexing still requires shared media;
- to better utilize scarce or expensive network resources.

2) What kind of multiplexing is used for public FM radio?

Public FM (Frequency Modulation) radio typically utilizes Frequency Division Multiplexing (FDM) to transmit multiple audio signals simultaneously over the same frequency band.

3) Which kind of multiplexing is used for GSM?

GSM (Global System for Mobile Communications) utilizes Time Division Multiple Access (TDMA) as its multiplexing technique. In TDMA, the available frequency band is divided into multiple time slots, and each user or communication channel is assigned one or more of these time slots for transmission.

4) What is the problem of multiplexing in time, what is its advantage?

The problem of multiplexing in time, particularly in Time Division Multiple Access (TDMA) systems like GSM, is that it requires precise synchronization among all users sharing the same frequency channel.

Advantages:

- Efficient spectrum utilization
- Flexibility
- Reduces interference
- Improved battery life

Modulation and Coding

1) How can we add information to a periodic signal? What are the choices?

We can add information to the periodic signal in the following ways:

Amplitude Modulation
Frequency Modulation
Phase Modulation

2) Why is amplitude modulation in its pure form rarely used in practice? When does it make sense? Where is the idea applied and what is the benefit of its application?

Amplitude Modulation in its pure form is used rarely because susceptibility to noise, bandwidth inefficiency, limited power efficiency.

It makes sense in the following categories:
Broadcasting and long-distance communication

The benefit of amplitude modulation is its compatibility with existing system and infrastructure.

3) What is the advantage of 16QAM over QPSK?

Increased spectral efficiency , Higher Data rates , Improved bandwidth utilization

4) What limits the use of higher order modulation schemes, as e.g. 16QAM or 64QAM?

Increased Susceptibility to Noise, Higher demands on channel conditions, Complexity and cost, bandwidth efficiency, Low SINR, Limit higher order modulation schemes

5) How can we optimize the throughput of a single transmitter-receiver pair? How can we optimize it under given bandwidth constraints as well as fixed signal and interference values?

Adaptive Modulation and Coding (AMC), Power control, Interference Mitigation, Spectrum Management, Advanced signal processing, Traffic optimization, Channel Equalization, Multiple Antenna technique, Packet aggregation, dynamic resource allocation, error control techniques

Spectrum Spreading

1) What is the benefit of spectrum spreading? What is needed for spreading?

Benefits of Spectrum Spreading:

- Resistance to interference
- Improved Security
- Multiple Access capability
- Robustness to Fading and Multipath

Components needed to achieve spectrum spreading:

- Spread Spectrum Modulation Technique
- Spreading code or hopping sequence
- Synchronization

2) What is the benefit of frequency hopping? What are its problems?

Benefits of frequency hopping:

- Resistance to Interference
- Improved security
- Frequency Diversity

Problems of Frequency Hopping:

- Synchronization complexity
- Limited spectral Efficiency
- Interference

Cellular Networks – Capacity and Interference

1) Spatial reuse in cellular systems: compare a reuse of 3 with a reuse of 9 with respect to capacity (data rate) and quality (error rate).

A spatial reuse factor of 3 generally results in higher capacity (data rate) and better quality (lower error rate) compared to a spatial reuse factor of 9. This is because lower reuse factors reduce interference between neighbouring cells, allowing for higher data rates and improved signal quality within each cell. However, the trade-off is that lower reuse factors require more spectrum resources and may lead to less efficient use of the available frequency bands

2) What is the effect of increasing the transmit power of all base stations in interference-limited systems?

Positive effects:

Improved coverage , Increased signal strength , Enhanced throughput

Negative effects:

Increased Interference, Reduced Capacity, Spectrum Congestion

3) How can we increase the system capacity in interference-limited cellular systems? And how not? Discuss the effects of different approaches.

In interference-limited cellular systems, increasing system capacity involves maximizing the number of users or data traffic that can be supported within the available spectrum while minimizing the impact of interference.

The following techniques increase system capacity in an interference-limited system:

Frequency reuse, interference mitigation techniques, Advanced modulation and coding, Multiuser MIMO(Multi input Multi Output), Dynamic spectrum access

Following approaches may not effectively increase system capacity:

Increasing transmit power, reducing cell size, Overlapping cells.

4) What are the results of adding more base stations to a given cellular network layout assuming that there is no extra spectrum available? What are the measures that need to be taken to handle this?

The results of adding more base stations to a given cellular network layout assuming that there is no extra spectrum available are following:

Positive effects:

Improved Coverage, Increased Capacity, enhanced quality of service

Negative Effects:

Interference, Increased costs, Complexity

The following measures need to be taken:

Interference Mitigation, Optimized Spectrum Management, Small cells Deployment, Advanced Antenna technologies, Network optimization, Backhaul capacity expansion

5) What are the effects of increasing the spatial separation, e.g. employing 6 instead of 3 sectors per base station?

Improved spatial reuse, Reduced interference, enhanced capacity, Improved coverage and quality, increased complexity, efficient resource allocation, cost considerations(Increase)

6) Discuss how direct communication between mobiles (D2D) is a solution to improve the network capacity of cellular networks. What are the gains of this?

The following are the gains:

Offloading Traffic from the Core Network

Localized Communication

Spectrum Efficiency

Improved coverage reliability

Enhanced throughput and data rates

Energy efficiency